CMPSC 475 Final Project Proposal

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1 Introduction

1.1 App Definition Statement

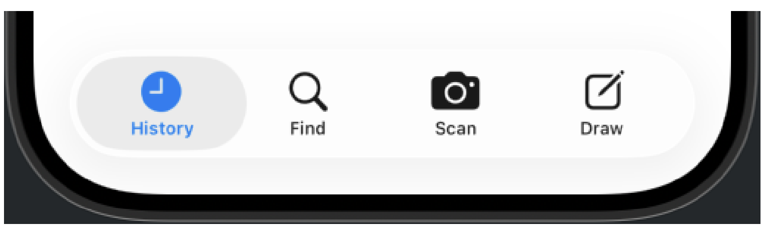
KanjiSearch is an app designed for beginner and intermediate Japanese language learners who are learning new vocabulary through textbooks or other print media. The purpose of this app is to efficiently look up and study Kanji characters found in physical media. The app provides three key search features: a search bar, a text-scanning camera, and a drawing frame. Both return the characters found, along with supplemental information, and all such "search sessions" are stored in a search history. Overall, the app is effective at alleviating the tediousness of looking up Kanji characters, so more time can be spent actually learning the language!

2 Features

This section outlines all the features that should be in the app for it to be considered "fully completed". However, as a base implementation, the most integral components of the app are the Draw Tab (2.3), the Scan Tab (2.4), and the Info Page (2.7). These three features are non-negotiable in fulfilling the app's main purpose: making Kanji character lookups more efficient and convenient.

2.1 Main Page

The Main Page will have four views: one for displaying previous searches (History) and three for using each of the different search features (Find, Scan, and Draw). By default, the app will open to the History Tab, and the tabs at the bottom of the screen can be used to switch to other views for searching.



The tabs on the Main Page

2.2 Find Tab

The Find Tab initially appears blank, save for a search bar at the top of the screen. As the user types in the search bar, found entries will display on a vertically scrollable list and update as the search changes.

The Tab View yielding three results from the current search string

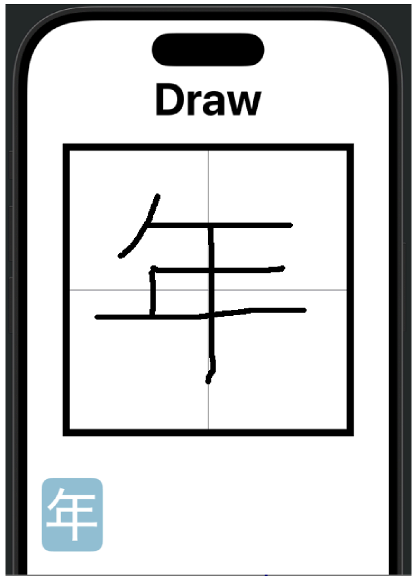
Each entry displays the character itself, its English translations, and its readings (On'yomi readings are written in Hiragana characters, while Kun'yomi readings are written in Katakana characters). Tapping on an entry leads to an Info Page for its respective Kanji character.

The user can begin searching with English letters, Japanese kana, or a Kanji character itself, and the query will adapt to which option they choose.

- Letters -> Filters by Translation
- Kana -> Filters by Reading
- Kanji -> Looks up the Character Itself

2.3 Draw Tab

The Draw Tab displays a blank canvas with gridlines that can be used to draw the Kanji character of interest. As more lines are added to the drawing, entries displaying Kanji that are most similar to the drawing populate below the canvas area. Tapping on one of these entries takes the user to an Info Page for its respective Kanji character.



The Draw Tab yielding a single result from the current drawing

2.4 Scan Tab

The Scan Tab prompts the user to take a photo of Japanese text using their device's camera. Given a valid photo, the user is taken to a Results Page displaying all of the Kanji found in the photo.

2.5 History Tab

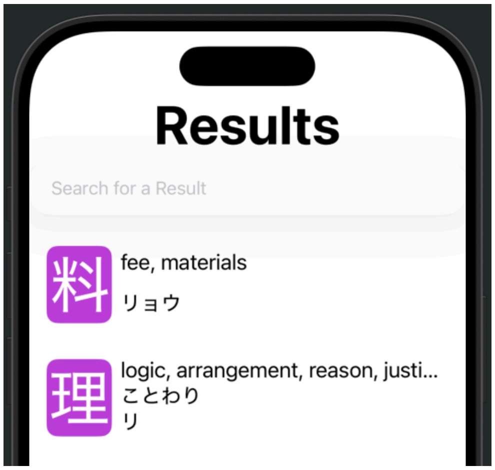
The History Tab shows a vertically scrollable list with entries representing past search sessions ordered by recency.

The History Tab with example search sessions of each type

Each entry shows the Kanji characters found in the session as well as which of the three search types was used. Tapping on an entry takes the user to its respective Info Page (for finds and draws) or Results Page (for scans).

2.6 Results Page

A Results Page contains a vertically scrollable list of the Kanji found during a single use of the text scanner. Tapping on an entry in the list takes the user to an Info Page for its respective Kanji character.



A Results Page with two results

The search bar at the top of the screen can be used to filter through results, and it adapts to different types of characters in the same manner as in the Find Tab.

2.7 Info Page

An Info Page will display a Kanji character alongside dictionary information. On top of translations and readings (as with other pages), it also displays the character's JLPT level as well as some compound vocabulary it is used in.

An Info Page displaying dictionary information for a singular Kanji character

3 Implementation & Technical Details

This section outlines some essential frameworks needed for the app and what features they pertain to.

3.1 Jisho API

The App's features that require dictionary information will be supported through Jisho API, a public, Python-based scraper of the Japanese dictionary website Jisho.org. A network call to this API will occur during the following scenarios:

1. The search bar in the Find Tab is updated.
2. The user needs to be taken to an Info Page or a Results Page.

3.2 UIKit

To access the device's camera for the Scan Tab, the app will include an embed to UIKit's "UIImagePickerController".

3.3 Vision/CoreML

Because the Vision framework's text recognition features are grounded upon instances of UIImage, it can act as the interpreter for the Scan Tab, in which it receives images from the UIKit embed. Using ImageRenderer, the canvas in the Draw Tab can also yield UIImage instances for Vision to interpret.

3.4 SwiftData

Local persistence for the History Tab shall be handled by SwiftData, with a container pertaining to a designated "search session model".

4 Stretch Goals

This section outlines extra features the app can include (or be updated with) given the time. They either enhance existing features or expand upon the use cases of the app.

4.1 Interactive Scan Results

With this upgrade, a Results Page yielded by a scan will have a second format for viewing Kanji entries: a digital recreation of the scanned text in which found Kanji are highlighted and, when tapped, take the user to the respective Kanji's Info Page.

4.2 VocabSearch

This extension allows both the Find Tab and Scan Tab to accept searches for Kanji-based vocabulary instead of just a Kanji character itself. Vocabulary terms would have their own Info and Results Pages. In addition, Kanji Info Pages will have a new button that, when tapped, takes the user to a Results Page displaying an extensive list of vocabulary that includes the character.

5 Schedule

- **Week 1: 4/5 - 4/11**
  This beginning week is meant for laying the groundwork of how the app should look and how data should be represented.
  - Create views for a general idea of how the app should be structured.
  - Structure models and provide samples of structs for use in views.

- **Week 2: 4/12 - 4/18**
  This week is allocated primarily for setting up the Jisho API, as it is imperative that the app can connect to the Japanese dictionary without fail.
  - Establish calls to Jisho API with network viewmodel and Python files.
  - Connect views to the network viewmodel.
  - Make "search" instances compatible with SwiftData
  - Begin structuring the Vision framework.

- **Week 3: 4/19 - 4/25**
  After the Jisho API is established, Vision and UIKit should be targeted next, completing the support needed for the Draw Tab and the Scan Tab.
  - Implement the drawing functionality of the Draw Tab's canvas.
  - Set up the UIKit controller for the Scan Tab.
  - Connect both the Draw Tab and Scan Tab to Vision text recognition requests

- **Week 4: 4/26 - 5/2**
  Given that the previous weeks went according to plan, this final week can be dedicated to small adjustments of the app and, if time allows, stretch goals.
    - Tweak views to be more visually appealing or user-friendly.
    - Add any notable quality of life features.
    - Implement stretch goals